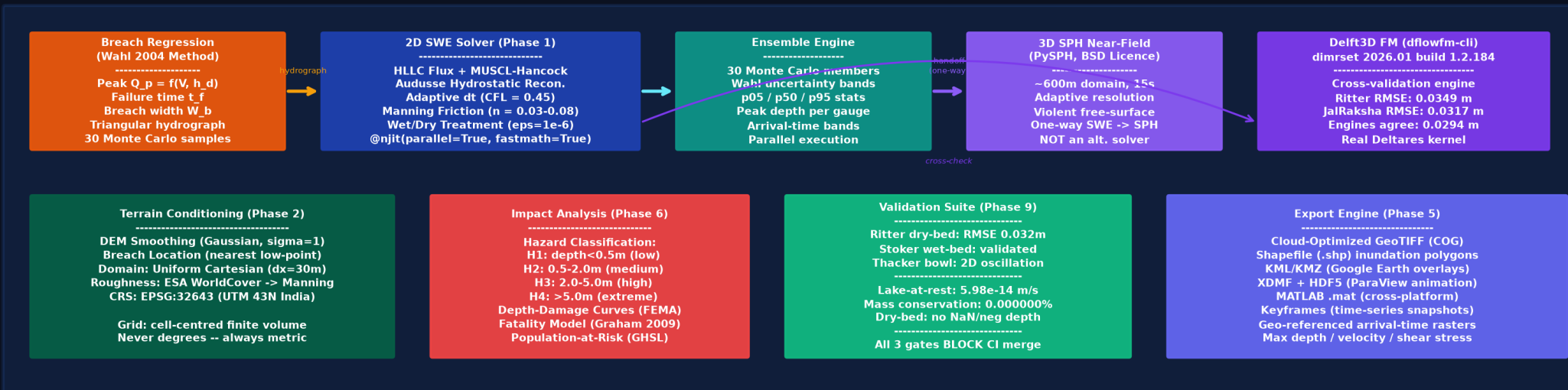


Multi-Layer Architecture for Dam-Break Inundation Modelling



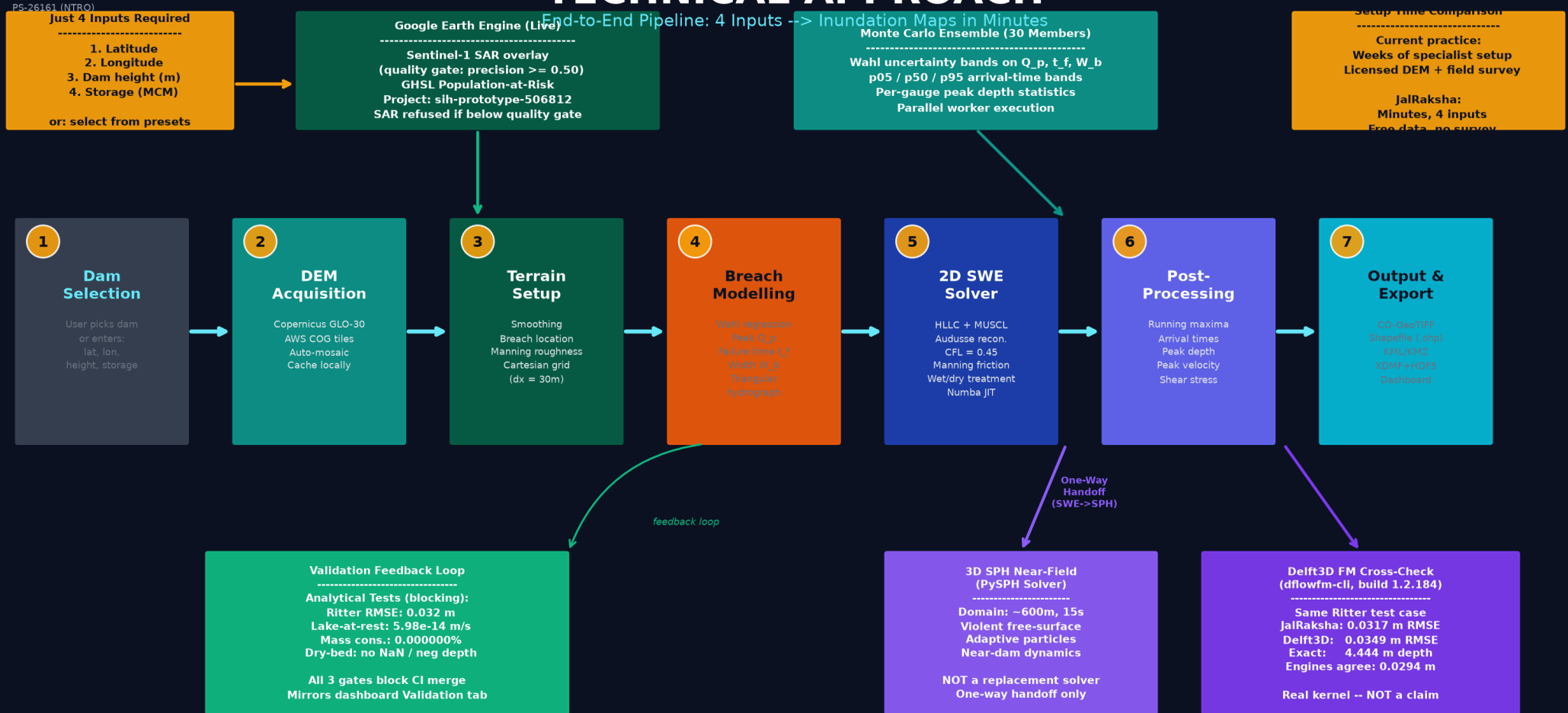
SOLVER ENGINE CORE



DATA SOURCES (Open Data Only -- No Commercial Licence)



End-to-End Pipeline: 4 Inputs --> Inundation Maps in Minutes



USER WORKFLOW DIAGRAM

End-to-End User Journey: Dam Selection --> Actionable Intelligence



TECHNOLOGY STACK

Components / Technology Stack to be Used

NUMERICAL SOLVERS

NumPy >= 1.24 Array computation

SciPy >= 1.10 Interpolation

Numba >= 0.57 JIT compilation

PySPH >= 0.9b0 SPH (BSD licence)

Matplotlib Visualization

GEOSPATIAL & DATA I/O

Rasterio >= 1.3 GeoTIFF read/write

GeoPandas >= 0.13 Vector data

Shapely >= 2.0 Geometry ops

xarray N-D arrays

netCDF4 / HDF5 Time-series I/O

PyProj >= 3.4 CRS transforms

BACKEND API

FastAPI Async REST API

uvicorn ASGI server

Pydantic Schema validation

SQLite Run database

Subprocess Worker pool

Click >= 8.1 CLI framework

FRONTEND DASHBOARD

React 18 Component UI

Vite 5 Build tool

Leaflet 1.9 2D map tiles

CesiumJS 1.144 3D globe

Recharts 3.10 Charts/graphs

Resium 1.25 React + Cesium

INFRA & DEVOPS

Docker Containerization

Docker Compose Multi-service

pytest + ruff Testing + lint

GitHub Actions CI/CD

Python 3.11+ Runtime

EXTERNAL DATA SERVICES (Open Data, Free, Cached Locally)

**Copernicus DEM
(AWS COGs)**

30m global
Free (ESA)

**Google Earth
Engine API**

Sentinel-1 SAR
GHSL population

ESA WorldCover

10m land-use
Manning lookup

**Delft3D FM
(dflowfm-cli)**

Deltares kernel
build 1.2.184

ParaView

3D scientific
visualization

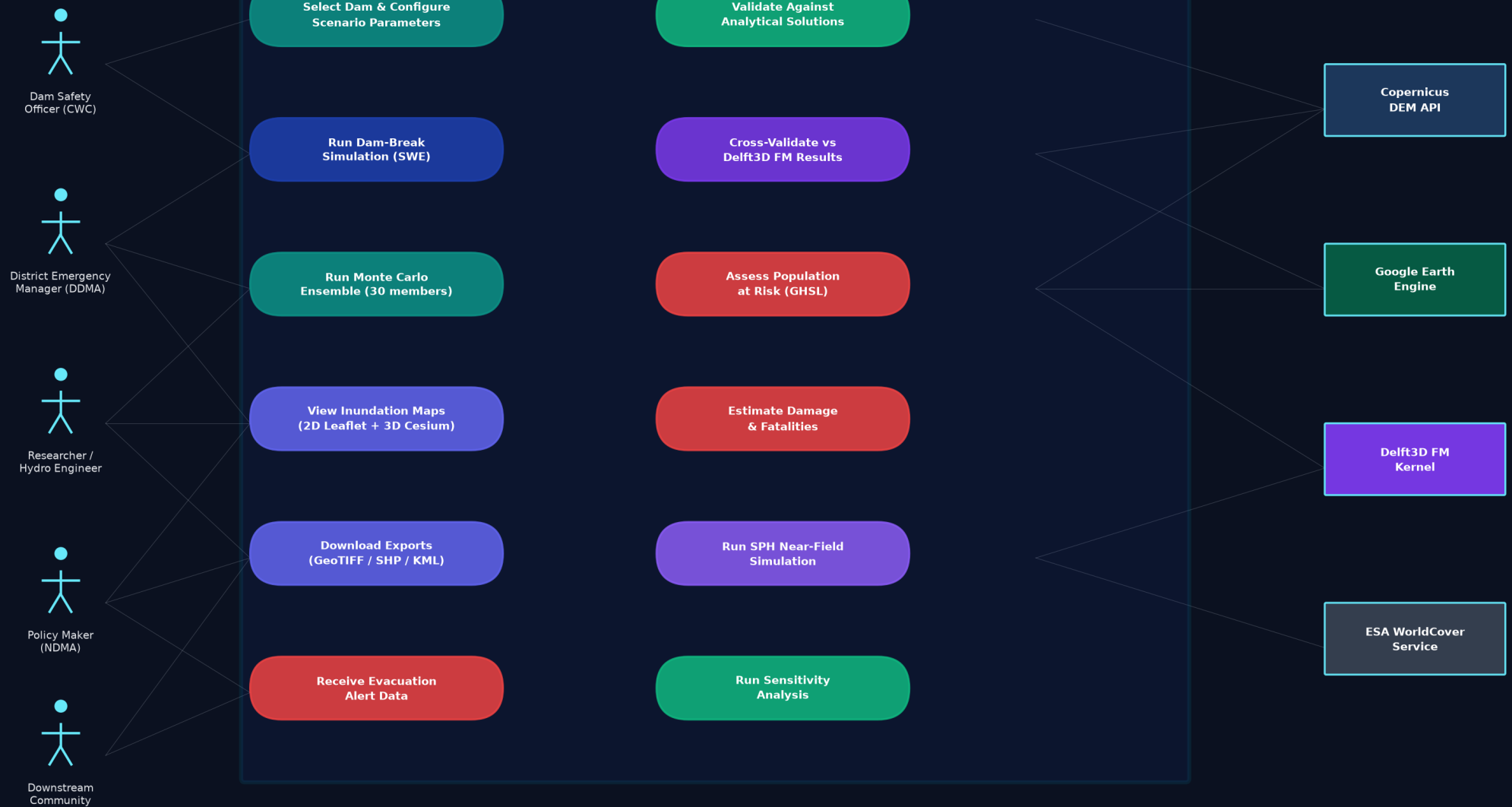
**CWC Dam
Register**

5000+ Indian
dams

USE CASE DIAGRAM

Actors & System Capabilities -- UML Style

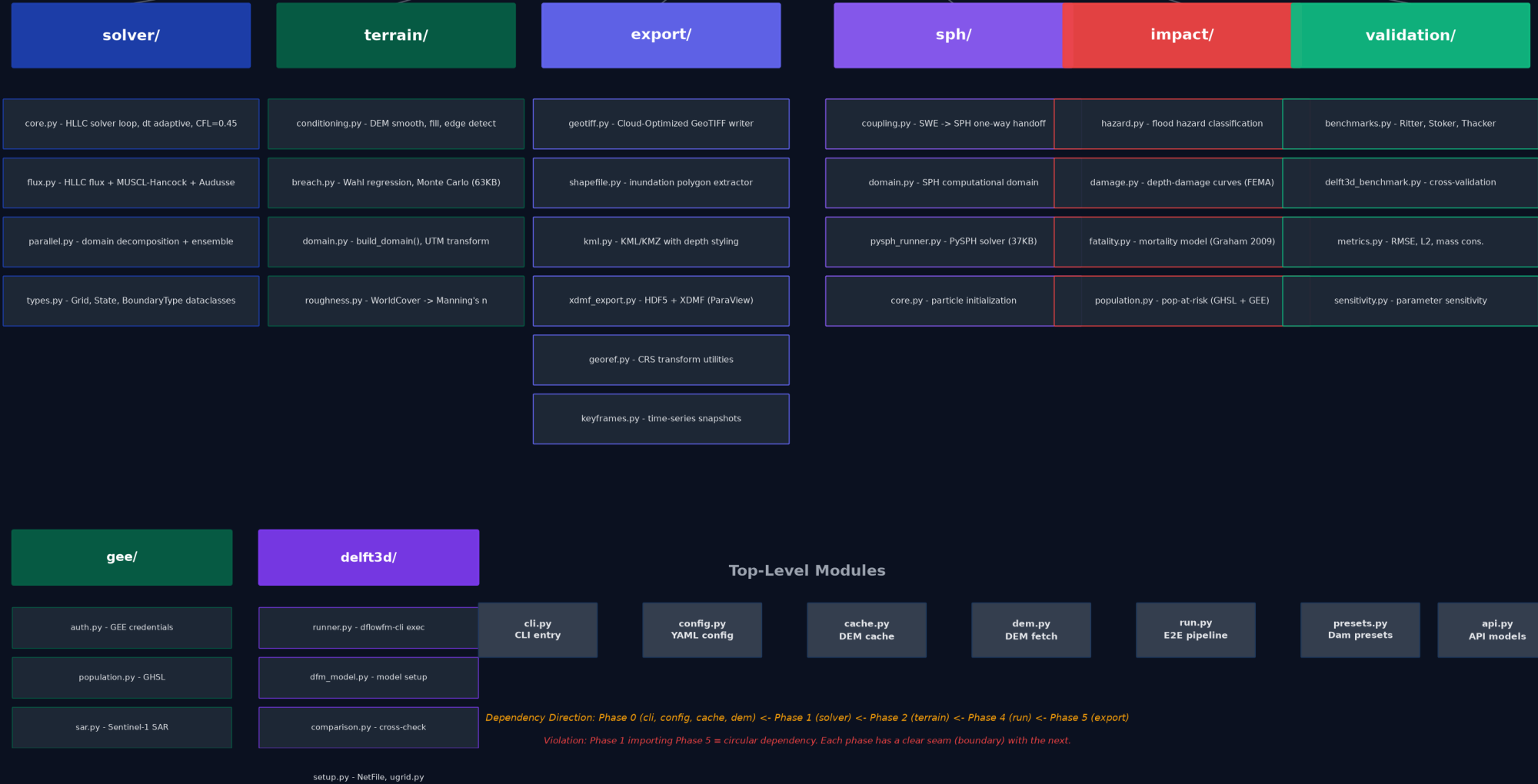
JalRaksha System Boundary



COMPONENT HIERARCHY

JalRaksha Package Structure & Module Dependencies

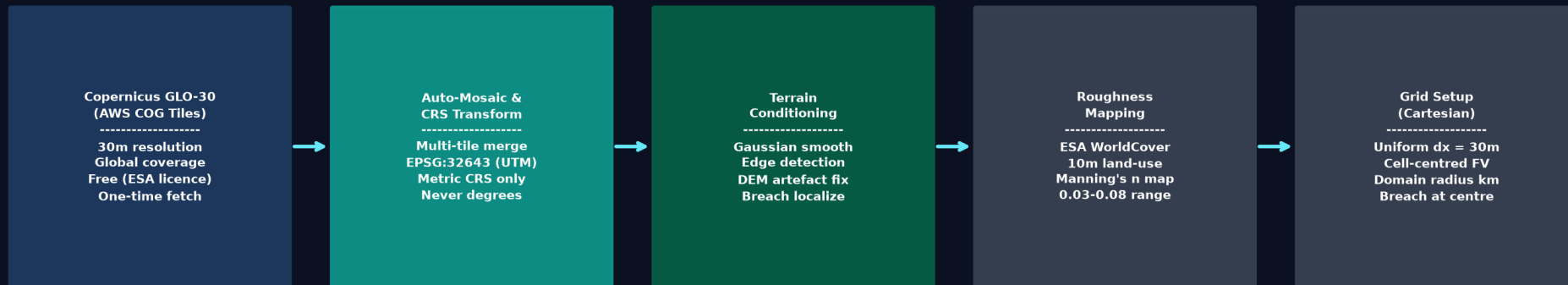
jalraksha/
(Python Package)



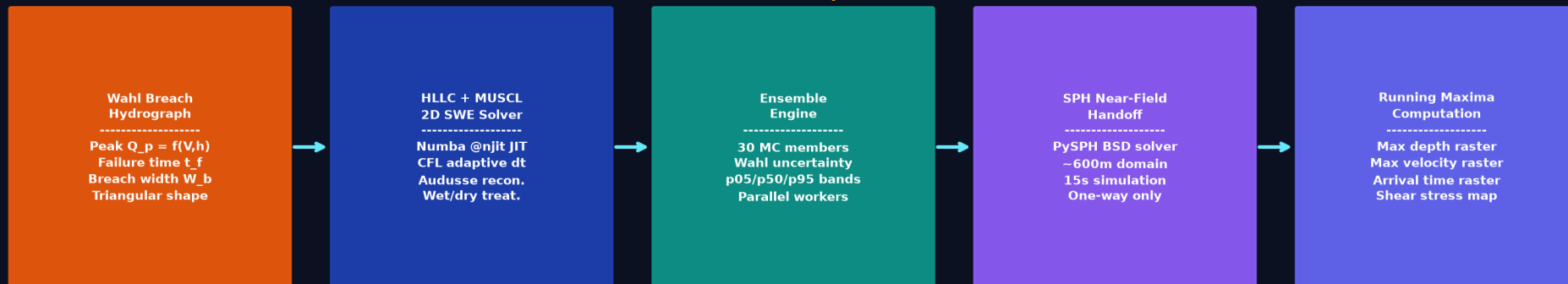
DATA PROCESSING PIPELINE

1. Data Acquisition & Terrain Setup

Three-Layer Architecture: Acquisition -> Processing -> Output



2. Hydrodynamic Solver & Ensemble Analysis



3. Impact Assessment, Export & Presentation



IMPACT AND BENEFITS

JalRaksha -- Positioned as Tier-1 Screening Tool for CWC Guidelines

Impacts:

- >> Rapid screening of 5,000+ CWC-registered dams for inundation risk in minutes, not weeks
- >> Open-data-only pipeline: no commercial licence, no field survey, no specialist required
- >> Ensemble uncertainty quantification (p05-p95) replacing single-point peak-flow estimates
- >> Cross-validated against Delft3D FM (real Deltares kernel): engines agree to 0.029 m RMSE
- >> Offline-first design: all data cached locally, demo-day network reliability assumed low

SOCIAL

- * Early evacuation warnings
- * Downstream community safety
- * CWC/NDMA risk visibility
- * SMS/Dashboard alerts
- * District-level planning

ECONOMIC

- * Rs.0 licence cost (open-source)
- * Minutes vs weeks per study
- * Prioritise high-risk dams first
- * Reduces specialist dependency
- * Budget-friendly for all states

ENVIRONMENTAL

- * Satellite-based land-use analysis
- * ESA WorldCover integration
- * Post-dam-break recovery plan
- * Downstream ecosystem mapping
- * Climate resilience screening

TECHNICAL

- * Validated HLLC solver (4 tests)
- * Delft3D FM cross-check (real kernel)
- * Monte Carlo ensemble (30 runs)
- * 3D SPH near-field handoff
- * Docker-based reproducibility

JalRaksha vs Current Practice:

Feature	Current Practice	JalRaksha
Setup time	Weeks of specialist setup	Minutes, 4 inputs
Data requirement	Licensed DEM + field survey	Free open data, no survey
Validation	No independent check	Scored vs Delft3D FM (real kernel)
Uncertainty	One number for peak flow	p05-p95 band (30 ensemble members)
SAR quality	Bad scenes used unfiltered	Refused at 0.50 precision gate
Network	Cloud-dependent	Offline-first (all cached locally)
Cost	Commercial software + survey	Open-source, MIT licence, Rs.0

TIER 1 : Analytical Exact Solutions (Blocking)

Ritter Dry-Bed
Dam-Break (1D)

h0 = 10m, dx = 10m, t = 40s
Exact: h = 4h0/9 = 4.444m
JalRaksha: 4.532m (+2.0%)
RMSE: 0.0317 m
3 boundary cells trimmed
Frictionless flat bed
STATUS: PASS

Stoker Wet-Bed
Dam-Break (1D)

1D shock-tube problem
Downstream water present
Shock + rarefaction wave
Jump conditions validated
Transonic expansion tested

STATUS: PASS

Thacker Parabolic
Bowl (2D)

2D oscillation in curved basin
Periodic analytical solution
Moving wet-dry shoreline
Convergence verified (dx halving)
Frequency matches exact

STATUS: PASS

TIER 2 : Blocking Correctness Gates (CI-Gated -- Must Pass Before Any PR Merge)

Lake-at-Rest

Max velocity < 0.1% threshold
Measured: 5.98e-14 m/s
Hydrostatic balance exact
Over arbitrary bathymetry

STATUS: PASS (13 orders below gate)

Mass Conservation

Volume loss < 0.1% threshold
Measured: 0.000000% loss
Over 1000 timesteps
Total volume tracked

STATUS: PASS (machine precision)

Dry-Bed Robustness

No NaN values anywhere
No negative depths
No division-by-zero
eps = 1e-6 threshold

STATUS: PASS (all cells stable)

TIER 3 : Delft3D FM Cross-Validation (dflowfm-cli, dimrset 2026.01, build 1.2.184)

Engine	RMSE vs Exact	Depth at Dam	Status
JalRaksha 2D SWE (built-in)	0.0317 m	4.532 m	PASS
Delft3D FM (dflowfm-cli)	0.0349 m	4.515 m	PASS
Exact (4h0/9)	--	4.444 m	Reference
Engine Agreement (mutual)	0.0294 m RMSE	--	VALIDATED

Engines agree to 0.0294 m RMSE

h0 = 10 m | t = 40 s
dx = 10 m | frictionless
3 boundary cells trimmed

DEPLOYMENT & INNOVATION

Offline-First Architecture + Docker Compose + What Makes JalRaksha Unique

Deployment Architecture



Innovation & Uniqueness

Dual-Engine Validation

First SIH project to cross-validate against a REAL Delft3D FM kernel (not claimed -- demonstrated)

Uncertainty Quantification

30-member Monte Carlo ensemble with p05-p95 uncertainty bands. No single-point estimates.

Open-Data Only

Zero commercial licences. Copernicus DEM + GEE + WorldCover. All free, all legal.

Domain Decomposition

2D SWE far-field + 3D SPH near-field. One-way handoff. Follows Maranzoni (2023).

SAR Quality Gate

Sentinel-1 SAR overlay REFUSED if precision < 0.50 vs JRC. No synthetic overlay ever shown.

Real Progress Tracking

progress_cb with phase string. 'Solving member 12/30' Not a frozen progress bar.

Roadmap: Done --> Now --> Next

DONE

Solver core (Phase 0-4)
Validation suite
Dashboard (8 tabs)
Delft3D cross-check
Ensemble engine

NOW

Impact analysis
SPH coupling polish
Docker hardening
Demo preparation
SIH 2026 submission

NEXT

Scale to 5000+ dams
NDMA integration
Mobile alerts
Tier-2 detailed studies
AWS/GCP deployment